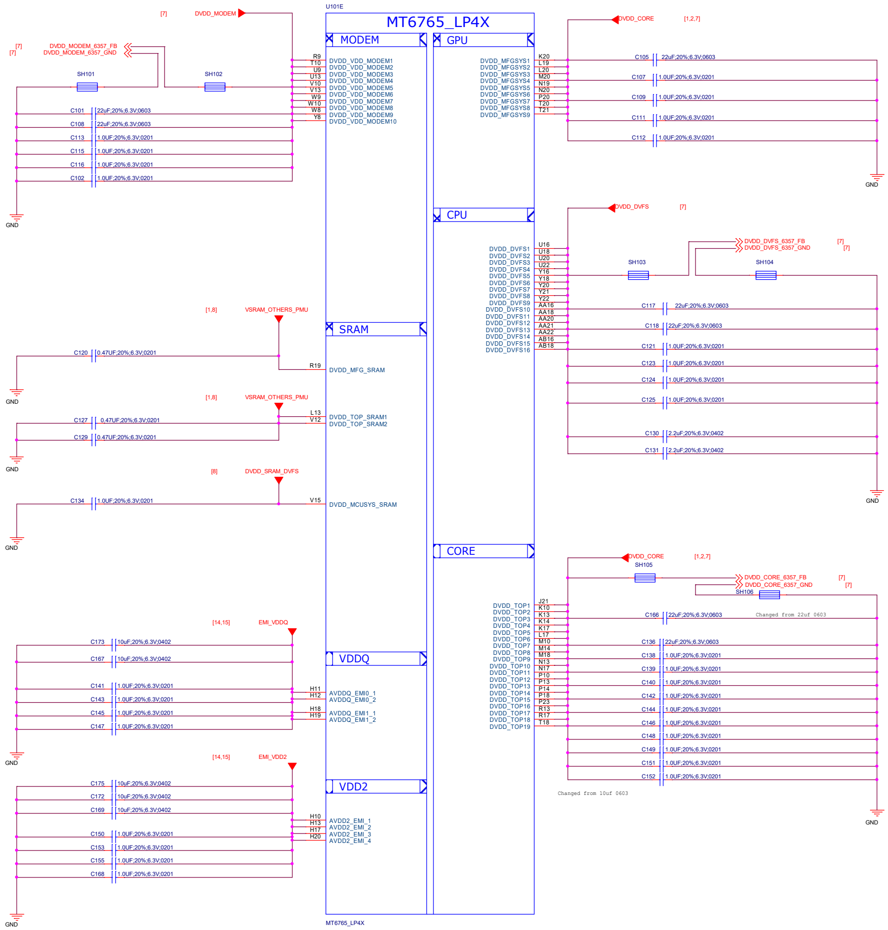
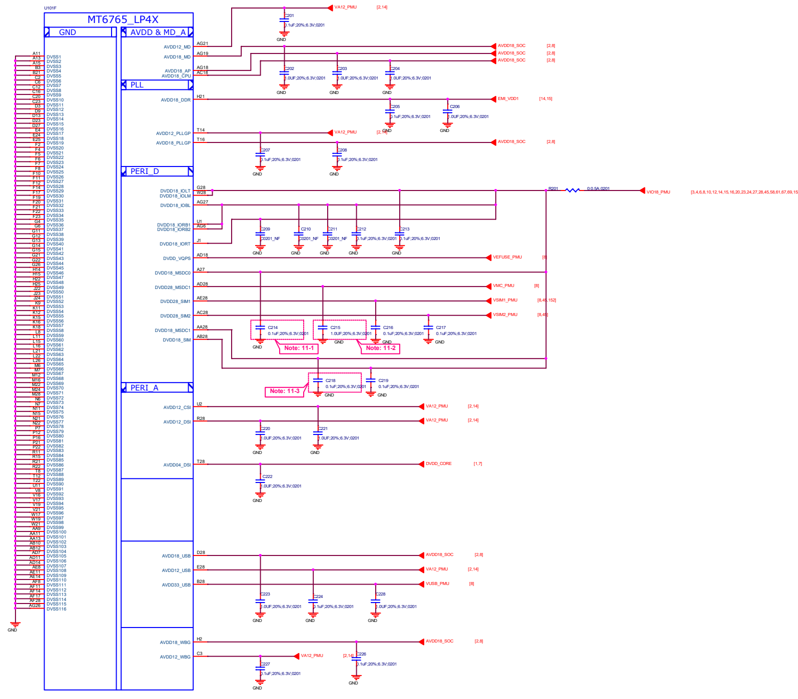


Title	01_BB_ POWER_PDN		REV: V10
DOCUMENT NO.:	Design Name		Size A2
DEPARTMENT:	WINGTECH-SH	DESIGNER:	LIUFENGLEI
Date:	Saturday, May 21, 2022	Sheet 1	of 53



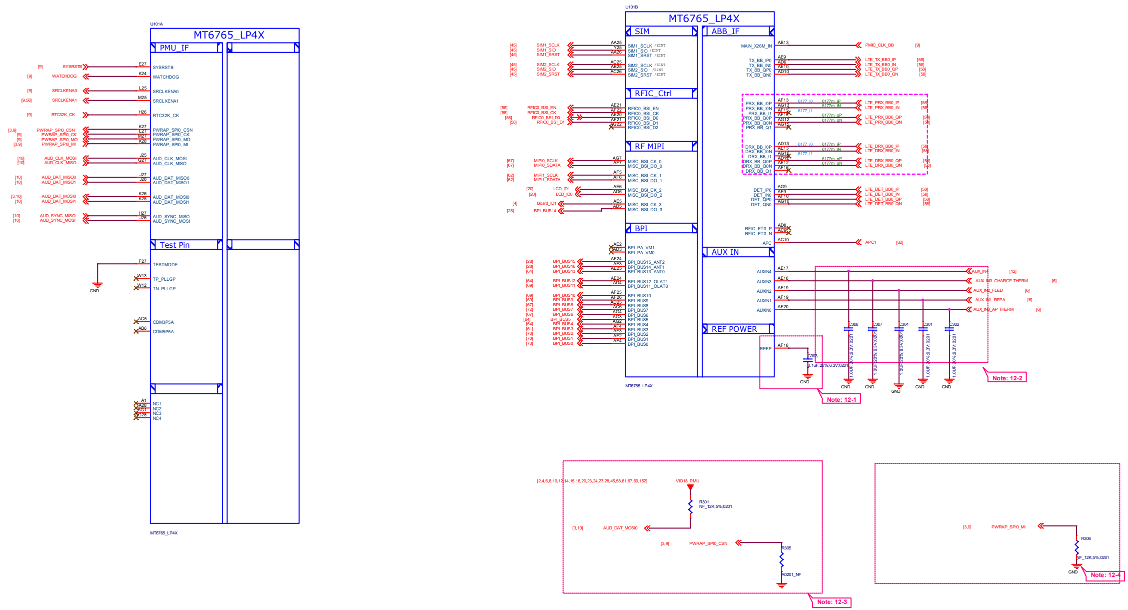
Schematic design notice of "11_BB_POWER_IO" page.

Note 11-1: C214 closed DVDD18_MSDC0 150mil

Note 11-2: C215 closed DVDD28_MSDC1 150mil

Note 11-3: C218 closed DVDD18_MSDC1 150mil

Title	02_BB_POWER_IO	REV:	V10
DOCUMENT NO.:	Design Name	Size	D
DEPARTMENT:	WINGTECH-SH	DESIGNER:	liufenglei
Date:	Saturday, May 21, 2022	Sheet	2 of 84



Schematic design notice of "12_BB_1" page.

Note 12-1: The de-coupling cap. for REFP (AF18 ball) have to be placed as close to BB as possible.

Note 12-2: To shunt a 1uF capacitor in the AUXIN ADC input to prevent noise coupling. It should be placed as close to BB as possible. Connect the unused AUX ADC input to GND.

Note 12-3: "PWRAP_SPI0_CSN" and "AUD_DAT_MOSI0" are bootstrap pin to select which interface will be the JTAG pin out.

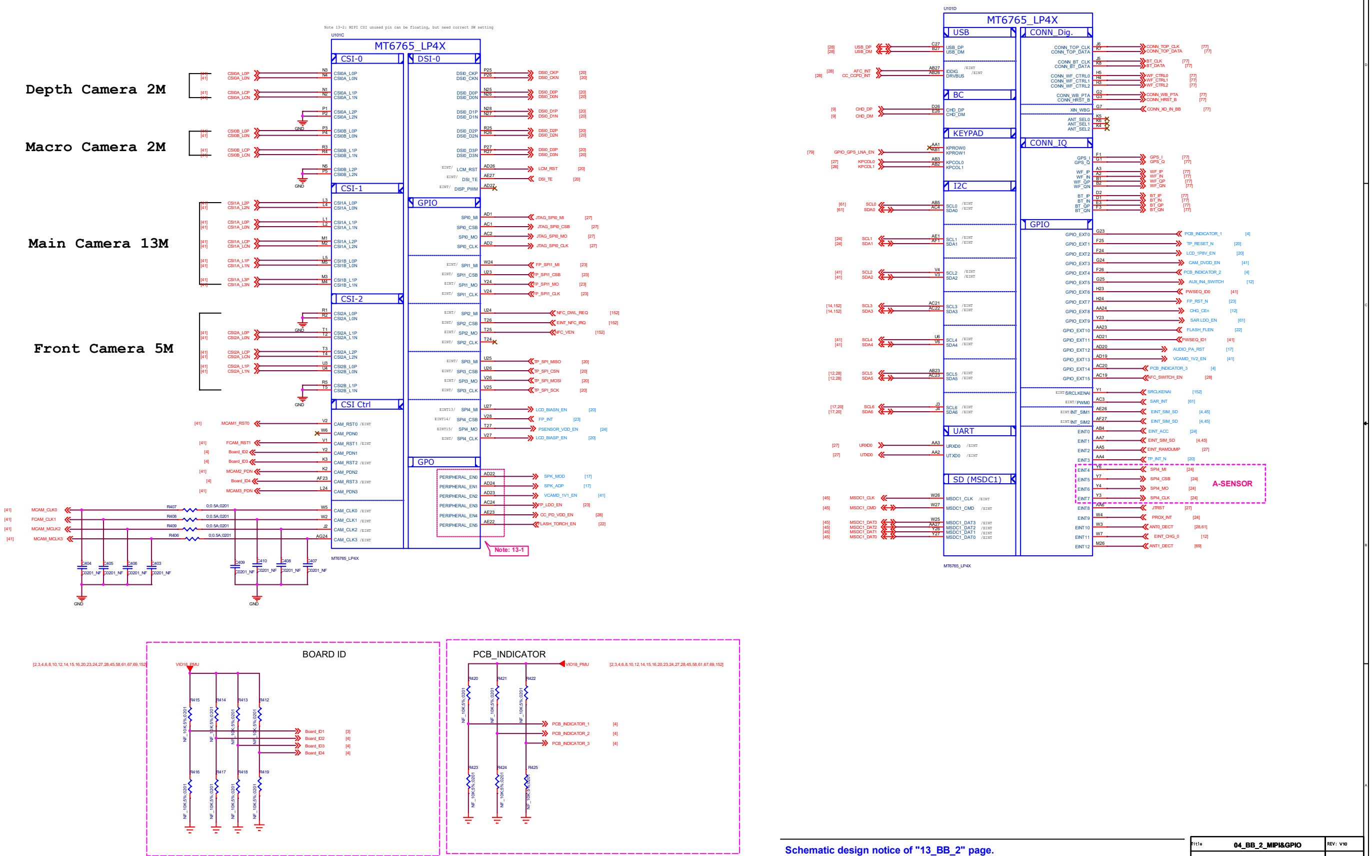
PWRAP_SPI0_CSN	AUD_DAT_MOSI0	JTAG Function	
default=PU	default=PD	AP_JTAG	MD_JTAG
HI	LO	N/A	N/A
HI	HI (by ext. PU)	SPI0+EINT8	SPI1+SPI3
LO (by ext. PD)	LO	SPI0+EINT8	N/A
LO (by ext. PD)	HI (by ext. PU)	MSDC1	N/A

Note 12-4: PWRAP_SPI0_MI / PWRAP_SPI0_MO is DDR type feature in bootstrap

PWRAP_SPI0_MI	PWRAP_SPI0_MO	Booting interface	
default=PU	default=PD	DDR	MSDC0 pin mux
LO (by ext. PD)	LO	LPDDR3	follow LPDDR3 Ref SCH.
LO (by ext. PD)	HI (by ext. PU)	N/A	N/A
HI	LO	LPDDR4X	follow LP4X/LP4 Ref SCH.
HI	HI (by ext. PU)	LPDDR4	follow LP4X/LP4 Ref SCH.

Note 12-5: Please set unused IQ pins in NC

Title	03_BB_1_RF&SIM_IF	REV:	V10
DOCUMENT NO.:	Design Name	Size	D
DEPARTMENT:	WINGTECH-SH	DESIGNER:	YL
Date:	Saturday, May 21, 2022	Sheet	3 of 46

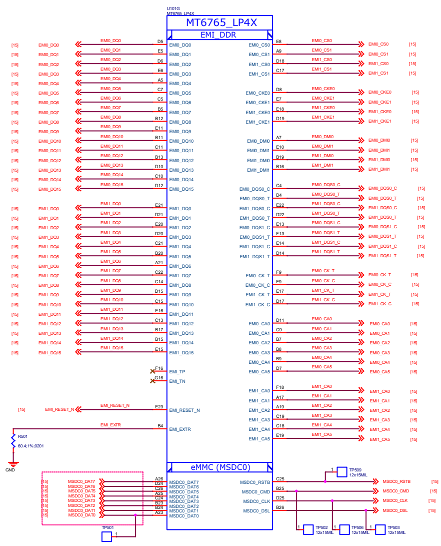


Schematic design notice of "13_BB_2" page.

Note 13-1: The enable pin of acoustic or optoelectronic devices (e.g. SPK AMP/Backlight/Charger OCP/OVP) suggest to use Peripheral_EN[0:5]
If use other GPIOs as enable pin, suggest to reserve 0201 NC to GND

Title	04_BB_2_MIPI&GPIO	REV:	V10
DOCUMENT NO.:	Design Name	Size	D
DEPARTMENT:	WINGTECH-SH	DESIGNER:	YL
Date:	Saturday, May 21, 2022	Sheet	4 of 46

WINGTECH



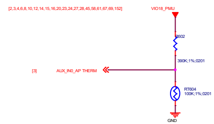
Schematic design notice of "14_BB_3" page.

Note 14-1: R501 please select 34.8 ohm (1%) resistor

Note 14-2: Please check eMCP LP3 and eMCP LP4X pin mux

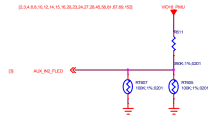
Title	05_BB_3_EMMC		REV: V10
DOCUMENT NO.:	Design Name		Size D
DEPARTMENT:	WINGTECH-SH	DESIGNER:	liufenglei
Date:	Saturday, May 21, 2022	Sheet 5 of 84	

The pull up ressister need to be 1% accuracy
NTC=100K

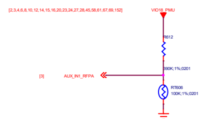


Thermistor to sense AP temperature

1. RT604must keep a distance about 6~8 mm away from AP and far from other heat sources 10 mm at least.
2. The distance is the shortest distance from package edge to edge.

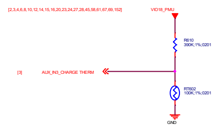


Thermistor to sense Flash LED



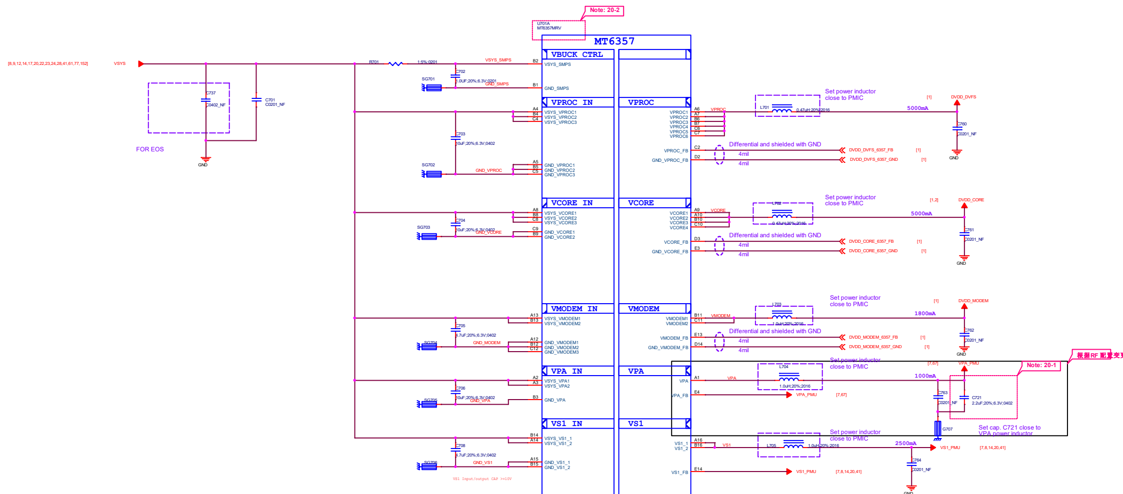
Thermistor to sense RF PA temperature

1. RT606 must close to LTE Band 7 PA or the hottest PA <2mm.
2. The distance is the shortest distance from package edge to edge.



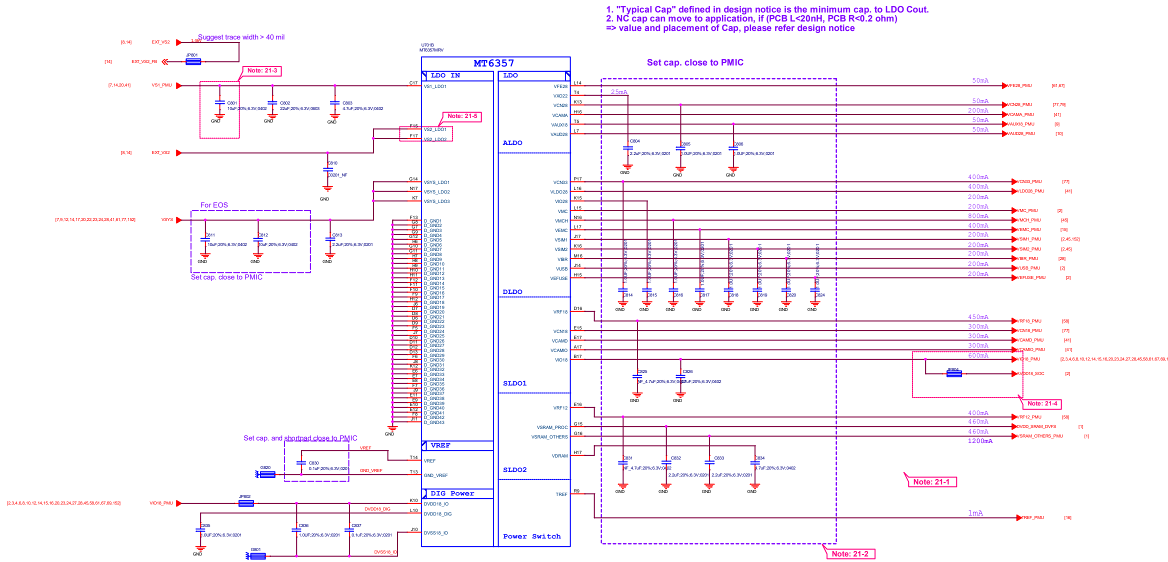
Thermistor to sense CHARGE temperature

Title	06_BB_AUXADC_Thermal	REV: V10	
DOCUMENT NO.:	Design Name	Size D	
DEPARTMENT:	WINGTECH-SH	DESIGNER:	YL
Date:	Saturday, May 21, 2022	Sheet 6 of 46	



Schematic design notice of "20_POWER_MT6357_Buck"	
Note 20-1:	C707, please choose 0402 size
Note 20-2:	For MT6765/62/61 platform, please only use MT6357CRV

Title	07_POWER_MT6357_Buck		REV: V10
DOCUMENT NO.:	Design Name		Size D
DEPARTMENT:	WINGTECH-SH	DESIGNER:	YL
WINGTECH			
Date:	Saturday, May 21, 2022	Sheet 7 of 46	



Schematic design notice of "21_POWER_MT6357_LDO"

Note 21-1: If these power trace can meet LDO layout constraint, these CAP can be NC or removed. Please refer to MT6357 design notice.

Note 21-2: Output cap range please follow MT6357CRV LDO design notice

Note 21-3: Ext Buck BOM option

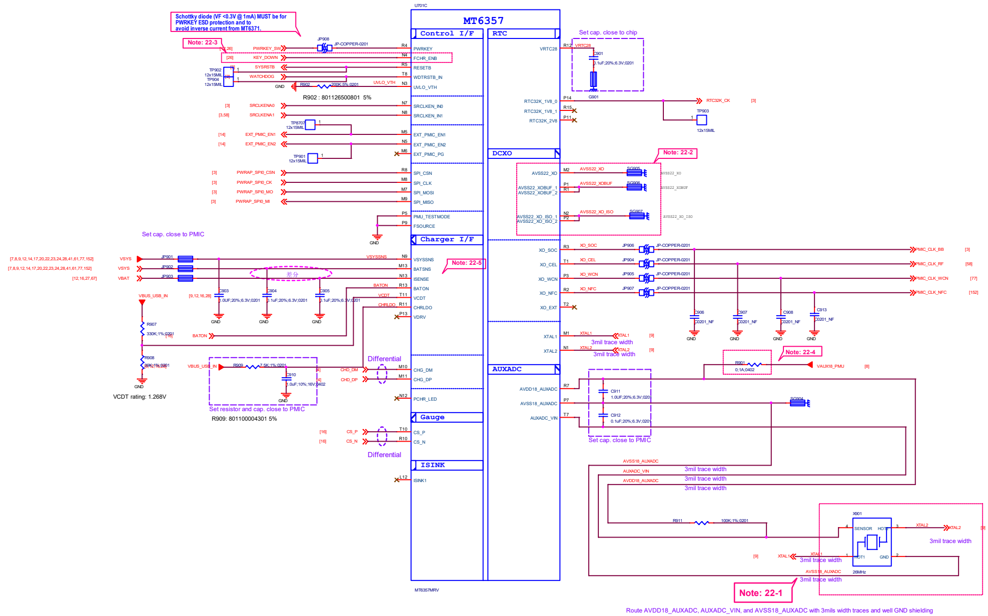
	Ext. buck option	
	w/ EXT VS2 Buck	w/o EXT VS2 Buck
C801	10uF	22uF

Note 21-4: Please set R803 and R803 close to C826, making star connection among VIO18_PMU, AVDD18_SOC, and EMI_VDD1 near to LDO cap. C826

Please also refer to MT6357 design notice for further detail design information

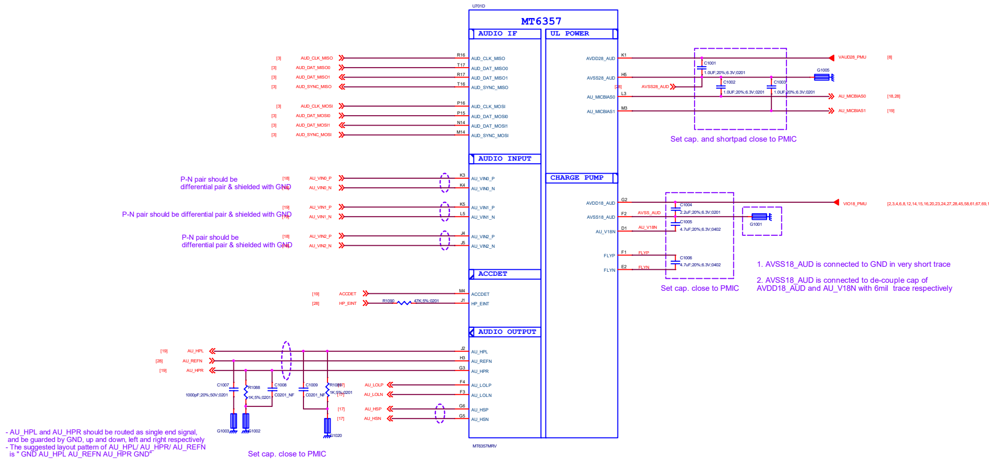
Note 21-5: Please connect VS2_LDO1(F15) to VS1_PMU if voltage applied to VCAMD(E17) >= 1.3 V

Title	08_POWER_MT6357_LDO	REV: V10	
DOCUMENT NO.:	Design Name	Size D	
DEPARTMENT:	WINGTECH-SH	DESIGNER:	YL
Date:	Saturday, May 21, 2022	Sheet 8	of 46

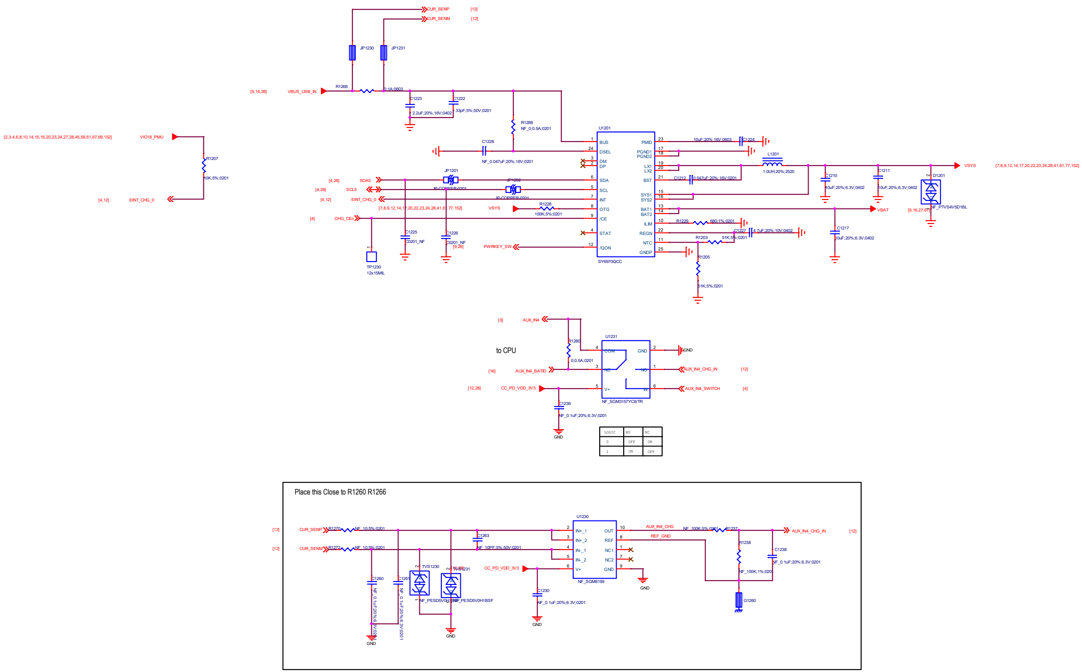


Schematic design notice of "22_POWER_MT6357-IF"	
Note 22-1:	Please implement 2520 & 2016 Size TMS PCB co-layout. Please refer to MT6762_MT6357 Co-Clock Design Notice for co-layout guide
Note 22-2:	1. Please Connect P1 and R1 ball first and then to GND 2. Please Connect P2 and N2 ball first and then to GND 3. Please connect DCXO GND to main GND by independent L1-2 GND via.; DO NOT connect it through L1 GND
Note 22-3:	Let floating if disable HOMEKEY function
Note 22-4:	Please follow MT6762_MT6357 Co-Clock Design Notice for Layout guide of VAUX18, then R8101 can use 0 ohm to replace BEAD. Please route VAUX18_PMU with well-ground sheilding if using 0 ohm to replace BEAD for AVDD18_AUXADC
Note 22-5:	Please connect to battery connector

Title	09_POWER_MT6357_IF	REV:	V10
DOCUMENT NO.:	Design Name	Size	D
DEPARTMENT:	WINGTECH-SH	DESIGNER:	YL
Date:	Saturday, May 21, 2022	Sheet	9 of 46



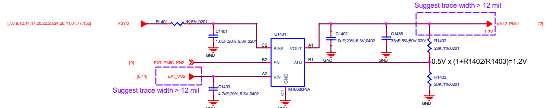
Title	10_POWER_MT6357_Audio	REV:	V10
DOCUMENT NO.:	Design Name	Size	D
DEPARTMENT:	WINGTECH-SH	DESIGNER:	YL
Date:	Saturday, May 21, 2022	Sheet	10 of 45



LOGIC	NO	NC
0	OFF	ON
1	ON	OFF

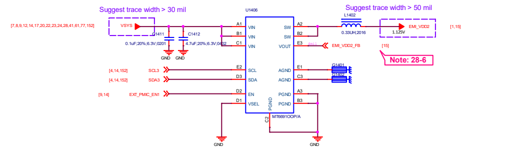
Title 12_POWER_SubPMIC-Charger + PP		REV: V10
DOCUMENT NO.: Design Name		Size D
DEPARTMENT: WINGTECH-SH	DESIGNER: YL	
Date: Saturday, May 21, 2022	Sheet 12 of 46	

LDO for VA12 Note: 28-1



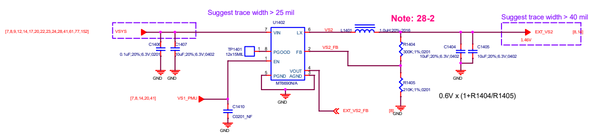
Ext. buck LPDDR4X/LPDDR4 VDRAM Note: 28-4

MT6691OOP/A / Ext. buck LP4X/LP4 VDRAM (VDD2)
I2C address: 0X57 (Write:0xAE, Read:0xAF)



Ext. Bulk for VS2

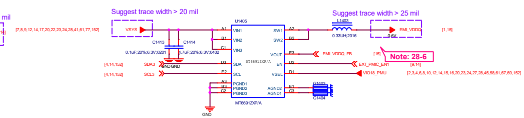
Note: 28-2



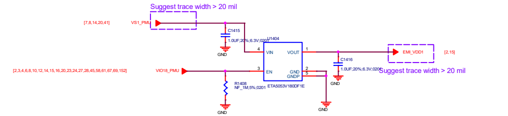
NOTE: Do not use VCAMD 1.3/1.5/1.8 when VS2 BUCK applied

Ext. buck LPDDR4X VDDQ Note: 28-4

MT6691ZXP/A / Ext. buck LP4X VDRAM(VDDQ)
I2C address: 0x50 (Write:0xA0, Read:0xA1)



LPDDR4X/LPDDR4 VDD1 1.8V LDO Note: 28-5

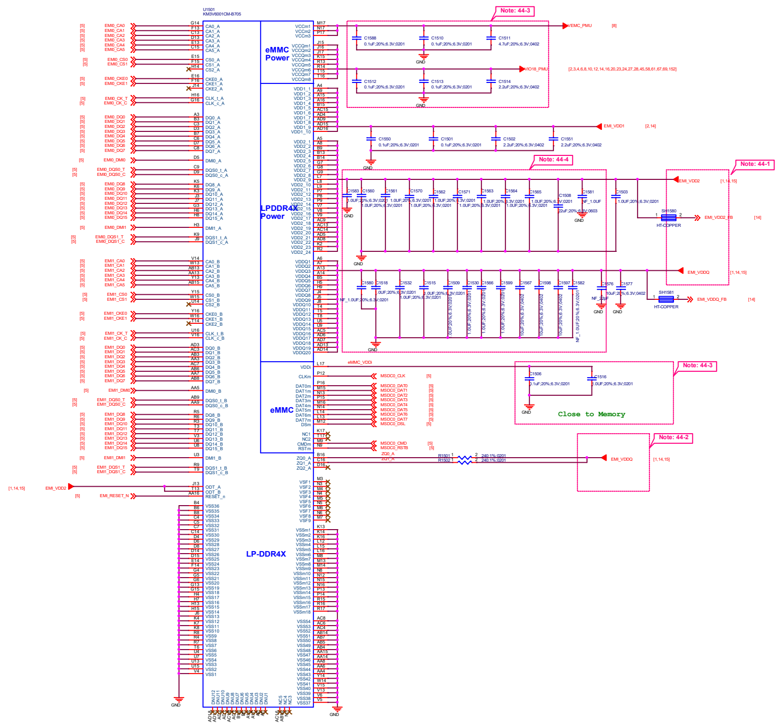


Schematic design notice of "28_POWER_ThirdParty-Power"

Note 28-1:	VA12 Layout placement please close to AP
Note 28-2:	VS2 Buck Layout placement please close to PMIC MT6357
Note 28-3:	VCN33 LDO Layout placement please close to MT6631
Note 28-4:	1.MT6691OOP/A Buck Layout Placement please close to LP4X/LP4 2.MT6691ZXP/A Buck Layout Placement please close to LP4X 3.If DRAM Application is LPDDR4 , MT6691ZXP/A NC
Note 28-5:	U2810 LDO Layout Placement Please close to LPDDR4X/LPDDR4 VDD1 power ball
Note 28-6:	For EMI_VDD2_FB and EMI_VDDQ_FB, please follow MMD rule

Title	14_POWER_ThirdParty_Powers	REV:	V10
DOCUMENT NO.:	Design Name	Size	D
DEPARTMENT:	WINGTECH-SH	DESIGNER:	YL
Date:	Saturday, May 21, 2022	Sheet	14 of 45

LPDDR4X

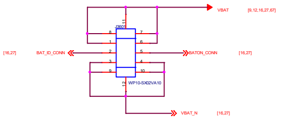


Schematic design notice of "44_Memory_eMMC_LPDDR4X_LPDDR4"	
Note 44-1:	Please refer to power supply related page select output voltage properly for LP4X/LP4
Note 44-2:	DRAM ZQx resistor = 240ohm (1%) that must be connected to VDDQ,
Note 44-3:	Please refer to eMCP vendor's datasheet or MTK common design notice to get the recommendation bypass cap. value for VCC/VCCQ/VDDI power domains of eMMC.
Note 44-4:	VDD2 VDDQ decoupling cap: closed to DRAM ball. For other cap for PMIC [>10uF, at PMIC page]: please also refer to MMD and layout guide for placement.

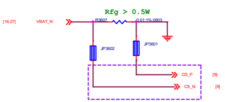
Note 44-5: Please check MT6765, MT6762 and MT6761's capacitor value.

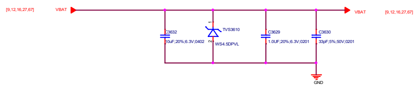
Title	15_Memory_eMMC_LPDDR4X	REV:	V10
DOCUMENT NO.:	Design Name	Size	D
DEPARTMENT:	WINGTECH-SH	DESIGNER:	liufenglei
Date:	Saturday, May 21, 2022	Sheet	15 of 84

Battery Connector



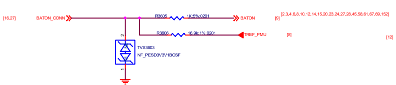
Guage Resistor





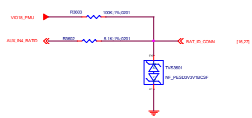
VBUS OVP

Battery NTC

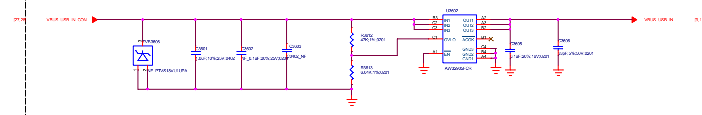


Battery pack is considered not present if BAT_ON is above 0.944*TREF

Battery ID



ADC 0.05-1.45 with 0.3 gap



Surge VBUS: ±100V with Overvoltage Protection up to 29V DC

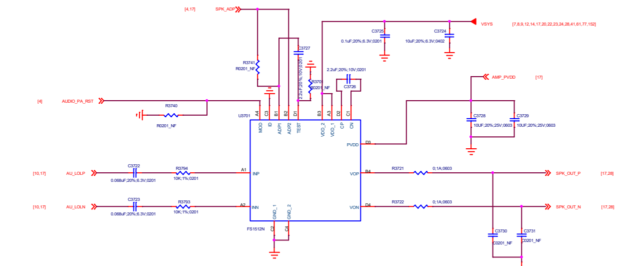
Vovp=10.5V when R3612=47K R3613=6.04K

Vovp=1.2*(1+ R3612/ R3613)

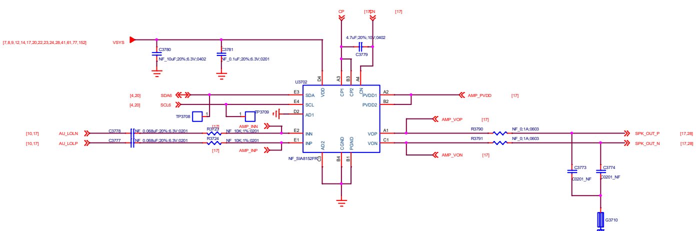
Title	36_Battery/USB IF	REV:	V10
DOCUMENT NO.:	Design Name	Size	D
DEPARTMENT:	WINGTECH-SH	DESIGNER:	YL
Date:	Saturday, May 21, 2022	Sheet 16 of	46

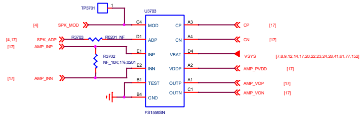
WINGTECH

Audio PA-Selection

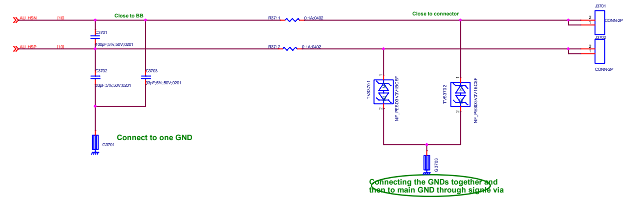


For FS1512N, R3741=0ohm,C3727=NC,R3701=0ohm
For SIA8102A & AW87318L, R3741=NC, C3727=2.2uF, R3701=NC.



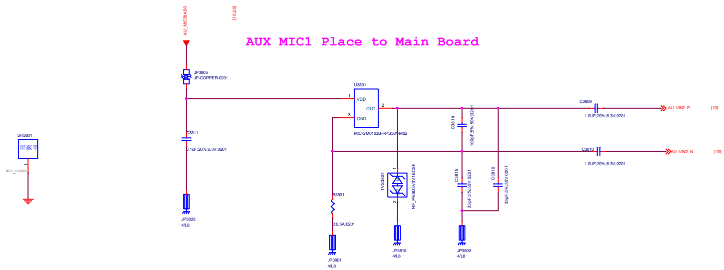


Receiver

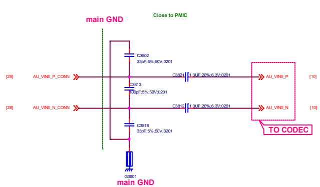


Title	37_SPK/Receiver/Vibrator	REV: V10
DOCUMENT NO.:	Design Name	Size D
DEPARTMENT: WINGTECH-SH	DESIGNER: YL	
Date: Saturday, May 21, 2022	Sheet 17 of 46	

AUX MIC

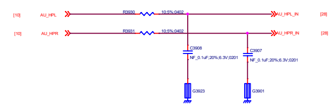


MAIN MIC

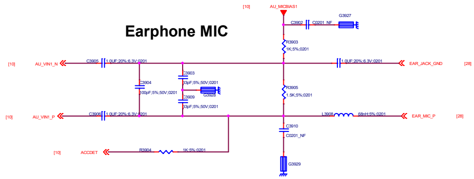


Title	38_MIC	REV: V10	
DOCUMENT NO.:	Design Name	Size	D
DEPARTMENT:	WINGTECH-SH	DESIGNER:	YL
Date:	Saturday, May 21, 2022	Sheet 18 of 46	

Earphone L/R



Earphone MIC



Title	39_Earphone	REV: V10	
DOCUMENT NO.:	Design Name	Size D	
DEPARTMENT:	WINGTECH-SH	DESIGNER: YL	
WINGTECH			
Date:	Saturday, May 21, 2022	Sheet 19 of 46	

Note 62-1: Part # of BEAD6202, BEAD6203, BEAD6204 and BEAD6205 needs changed to "BLM18BD102SN1" for high THD performance (-90dB) but this BOM change will results in FM RSSI 10dB degraded .

Note 62-2: Reserved Cap for CS/RS test, please double check multi-key function when used

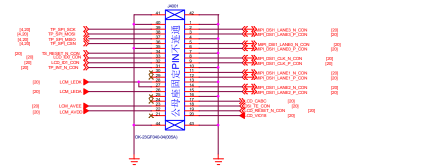
Note 62-3:

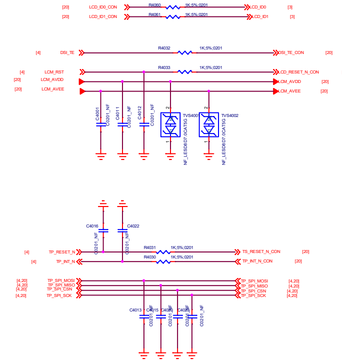
	Earphone Jack @ Main board	Earphone Jack @ Sub board
R6513	0 ohm	100nF

Note 62-4: Please Select ACC Mode for Operator Project to Pass Electrical MOS Test;
More Information refer to Audio/Speech Design Notice

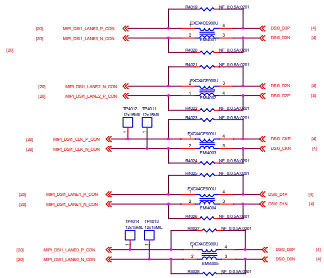
Note 62-5: Please select R6231 with 0402 size

LCM BTB

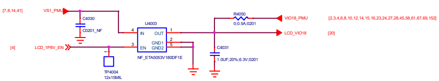




Common Mode Filter

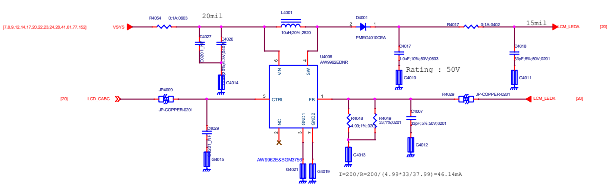


Reserve for ESD

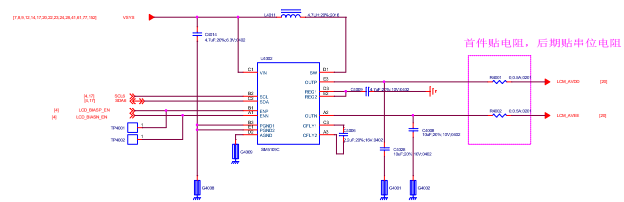


LCD Backlight LED Driver

LCD Backlight LED Driver



LCD Gate Drive



Title	40_LCD/CTP IF	REV:	V10
DOCUMENT NO.:	Design Name	Size	D
DEPARTMENT:	WINGTECH-SH	DESIGNER:	YL
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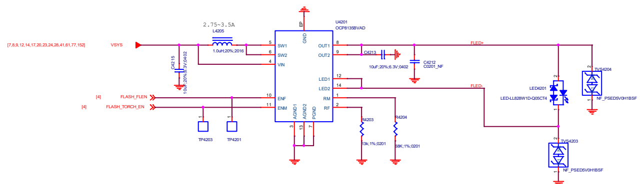
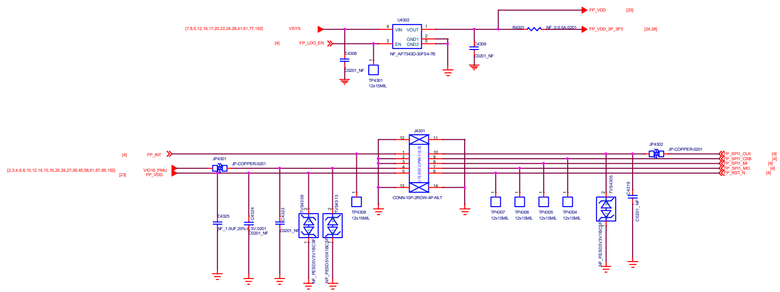
Flash driver

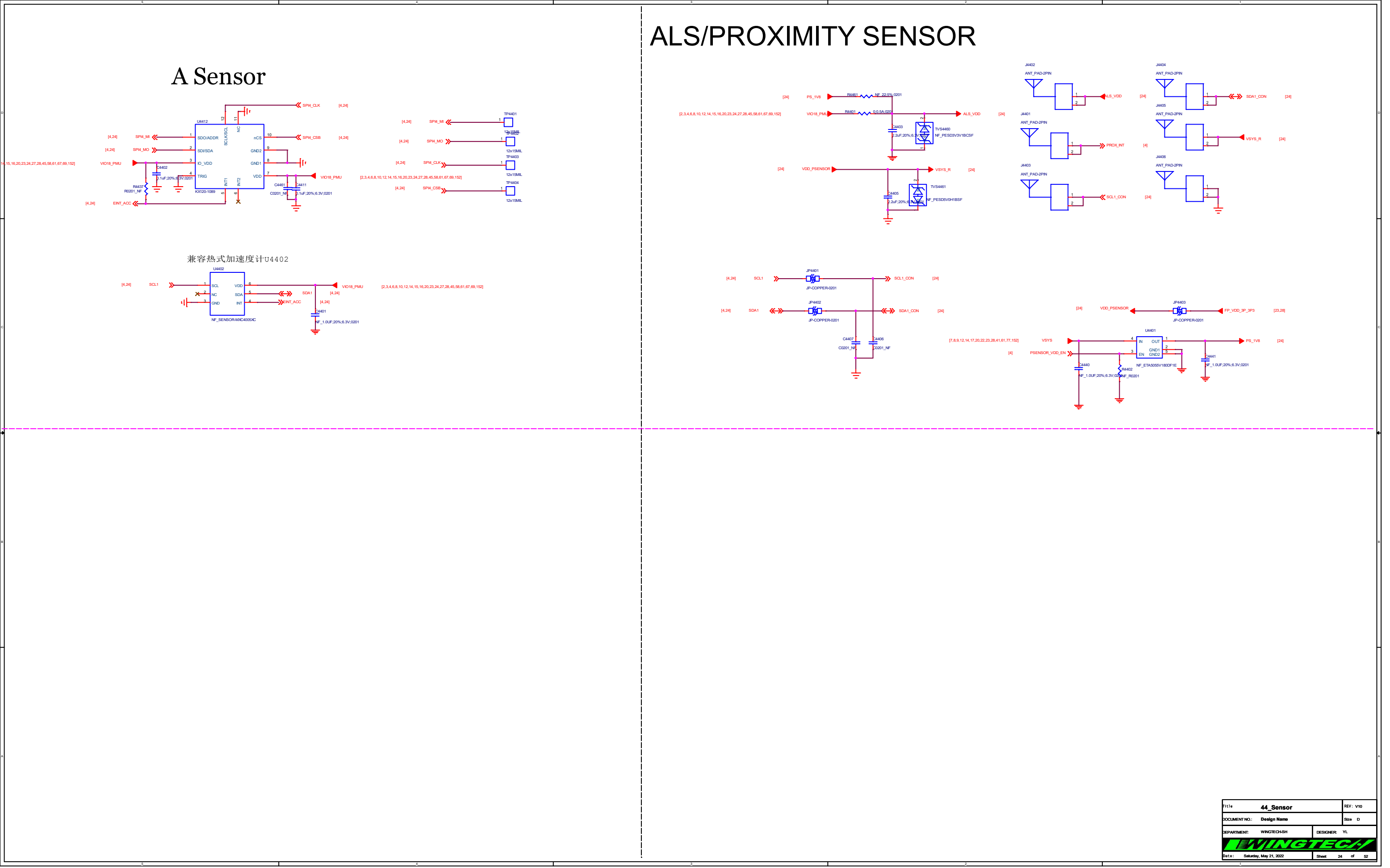


Title	42_Flash/RGB	REV: V10	
DOCUMENT NO.:	Design Name	Size D	
DEPARTMENT:	WINGTECH-SH	DESIGNER:	YL
Date:	Saturday, May 21, 2022	Sheet 22	of 46

Fingerprint

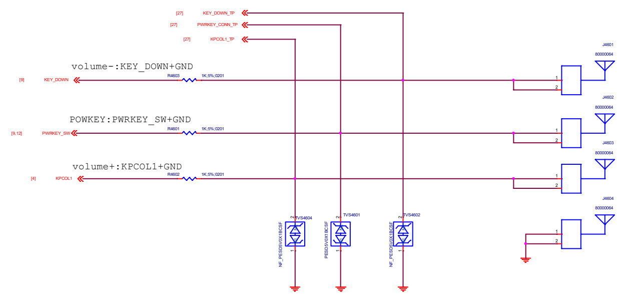


Title	43.Fingerprint	REV:	V10
DOCUMENT NO.:	Design Name	Size	D
DEPARTMENT:	WINGTECH-SH	DESIGNER:	YL
Date:	Saturday, May 21, 2022	Sheet 23	of 46

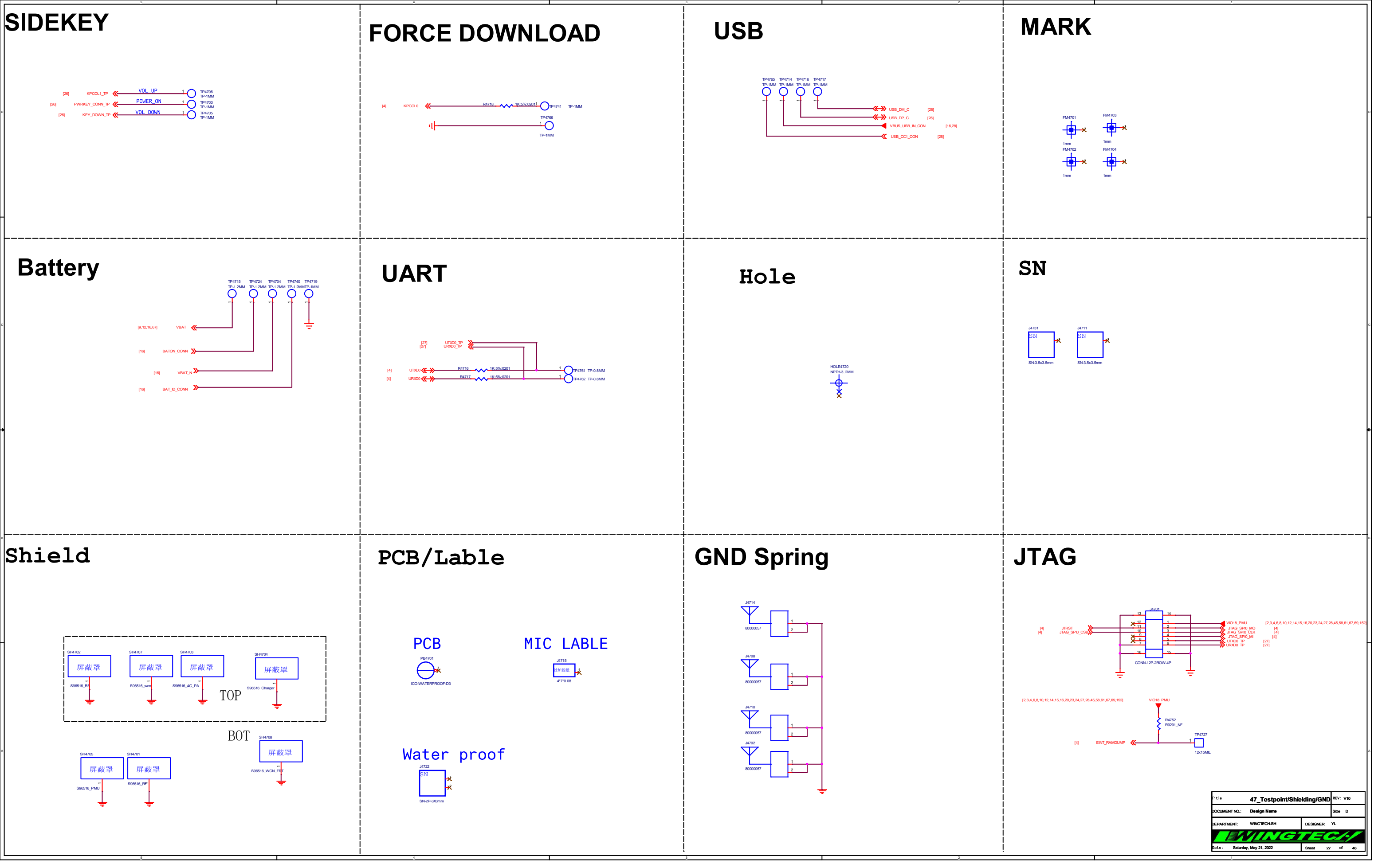

A Sensor
ALS/PROXIMITY SENSOR
U4412
SDO/ADDR
SDI/SDA
IO_VDD
TRIG
SCLK/SCL
NC
nCS
GND2
GND1
VDD
INT1
INT2
KX120-1089
SPI4_CLK
SPI4_MI
SPI4_MO
SPI4_CSB
VIO18_PMU
EINT_ACC
R4437
R0201_NF
C4402
C4461
C0201_NF
C4411
TP4401
TP4402
TP4403
TP4404
12x15MIL
兼容热式加速度计U4402
U4402
SCL
NC
GND
VDD
SDA
INT
SCL1
SDA1
NF_SENSOR-MXC4005XC
C4401
NF_1.0UF;20%;6.3V;0201
[2,3,4,6,8,10,12,14,15,16,20,23,24,27,28,45,58,61,67,69,152]
PS_1V8
R4461
R4401
VIO18_PMU
ALS_VDD
C4403
TVS4460
NF_PESD3V3V1BCSF
VDD_PSENSOR
VSYS_R
C4405
TVS4461
NF_PESD5V0H1BSF
J4401
J4402
J4403
J4404
J4405
J4406
ANT_PAD-2PIN
PROX_INT
SCL1_CON
SDA1_CON
JP4401
JP4402
JP4403
JP-COPPER-0201
C4407
C4406
C0201_NF
FP_VDD_3P_3P3
U4401
IN
EN
OUT
GND1
GND2
VSYS
PSENSOR_VDD_EN
C4440
C4441
R4402
NF_R0201
NF_ETA5055V180DF1E
[7,8,9,12,14,17,20,22,23,28,41,61,77,152]
NF_1.0UF;20%;6.3V;0201
Title 44_Sensor
REV: V10
DOCUMENT NO.: Design Name
Size D
DEPARTMENT: WINGTECH-SH
DESIGNER: YL
WINGTECH
Date: Saturday, May 21, 2022
Sheet 24 of 52

Power Key / Key Pad

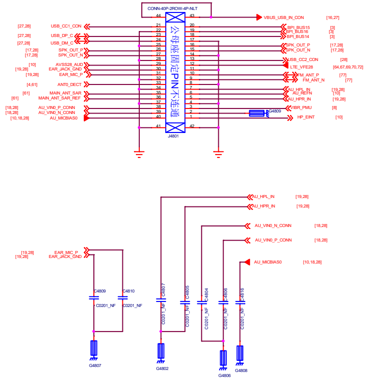
DO NOT put pull-up resistor on PWRKEY



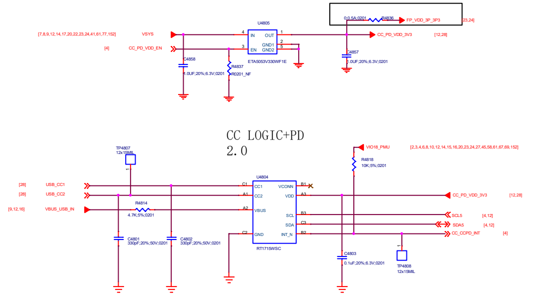
Title	46_Sidekey	REV:	V10
DOCUMENT NO.:	Design Name	Size	D
DEPARTMENT:	WINGTECH-SH	DESIGNER:	YL
Date:	Saturday, May 21, 2022	Sheet 26	of 46


SIDEKEY
FORCE DOWNLOAD
USB
MARK
Battery
UART
Hole
SN
Shield
PCB/Lable
GND Spring
JTAG
VOL_UP
POWER_ON
VOL_DOWN
KPCOL1_TP
PWRKEY_CONN_TP
KEY_DOWN_TP
KPCOL0
USB_DM_C
USB_DP_C
VBUS_USB_IN_CON
USB_CC1_CON
VBAT
BATON_CONN
VBAT_N
BAT_ID_CONN
UTXD0_TP
URXD0_TP
UTXD0
URXD0
HOLE4720
NPTH-3_2MM
SN-3.5x3.5mm
屏蔽罩
TOP
BOT
PCB
MIC LABLE
Water proof
ICO-WATERPROOF-D3
SN-2P-3X3mm
CONN-12P-2ROW-4P
VIO18_PMU
JTRST
JTAG_SPI0_CSB
JTAG_SPI0_MO
JTAG_SPI0_CLK
JTAG_SPI0_MI
EINT_RAMDUMP
R0201_NF
47_Testpoint/Shielding/GND
Design Name
WINGTECH-SH
YL
V10
Saturday, May 21, 2022
Sheet 27 of 46
WINGTECH

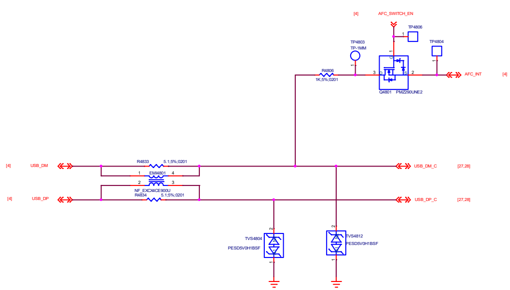
SUB CONNECTOR



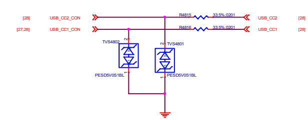
USB PD



USB_DP DM

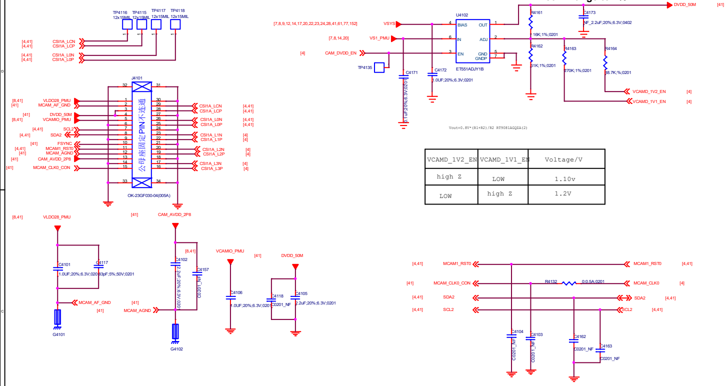


USB CC



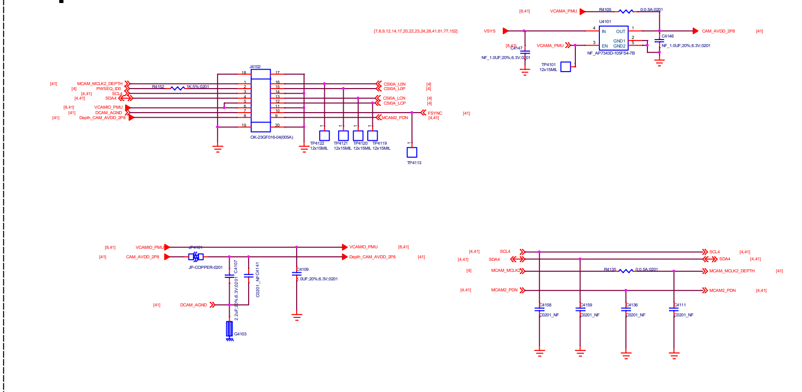
Title	48_Sub PCB IF	REV:	V10
DOCUMENT NO.:	Design Name	Size	D
DEPARTMENT:	WINGTECH-SH	DESIGNER:	YL
WINGTECH			
Date:	Saturday, May 21, 2022	Sheet	28 of 45

Main Camera 50M

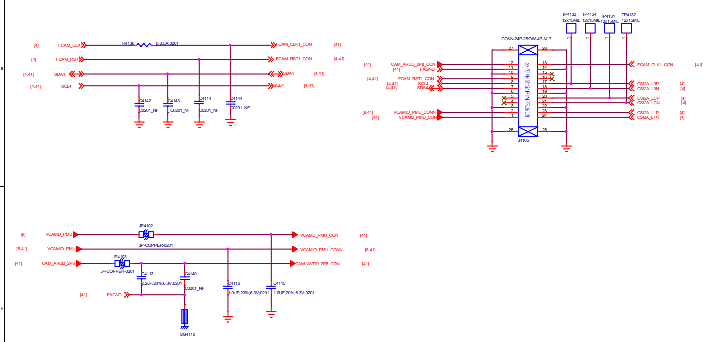


VCAMD_1V2_EN	VCAMD_1V1_EN	Voltage/V
high Z	LOW	1.10v
LOW	high Z	1.2V

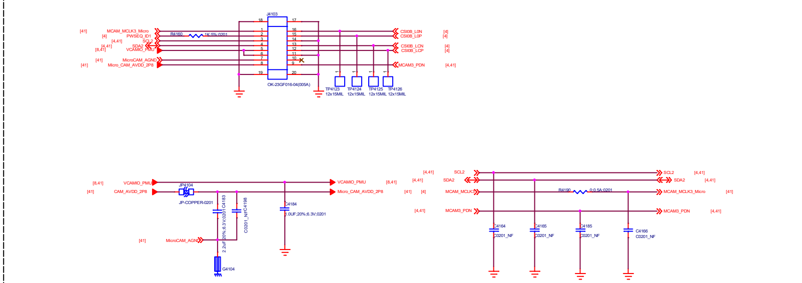
Depth Camera 2M



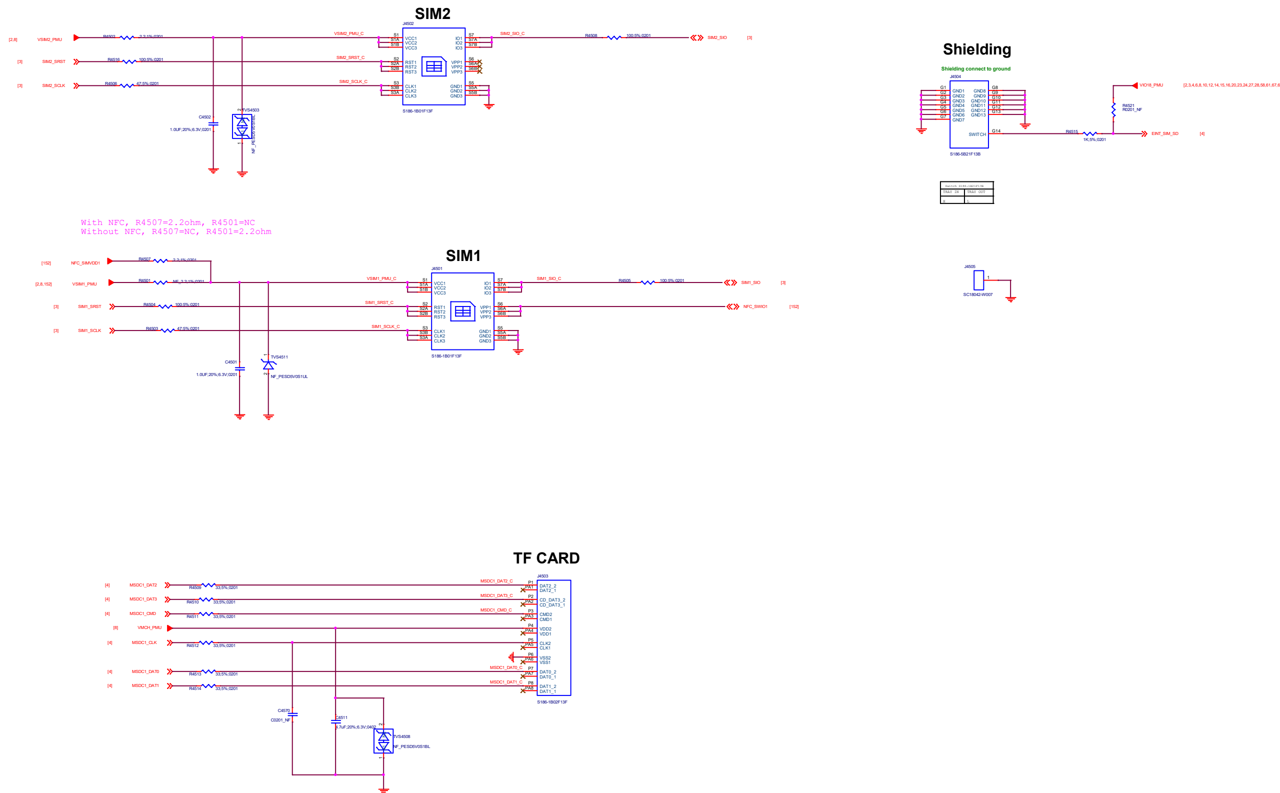
Front Camera 5M



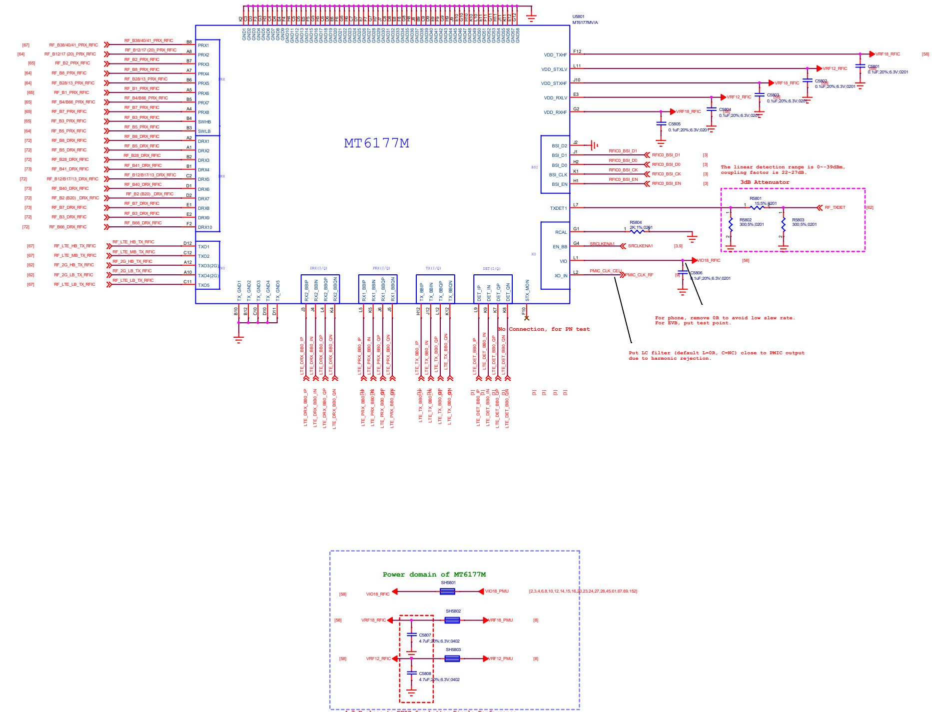
Micro Camera 2M



Title	41_Camera IF	REV:	V10
DOCUMENT NO.:	Design Name	Size	D
DEPARTMENT:	WINGTECH-SH	DESIGNER:	liufenglei
Date:	Saturday, May 21, 2022	Sheet	41 of 84

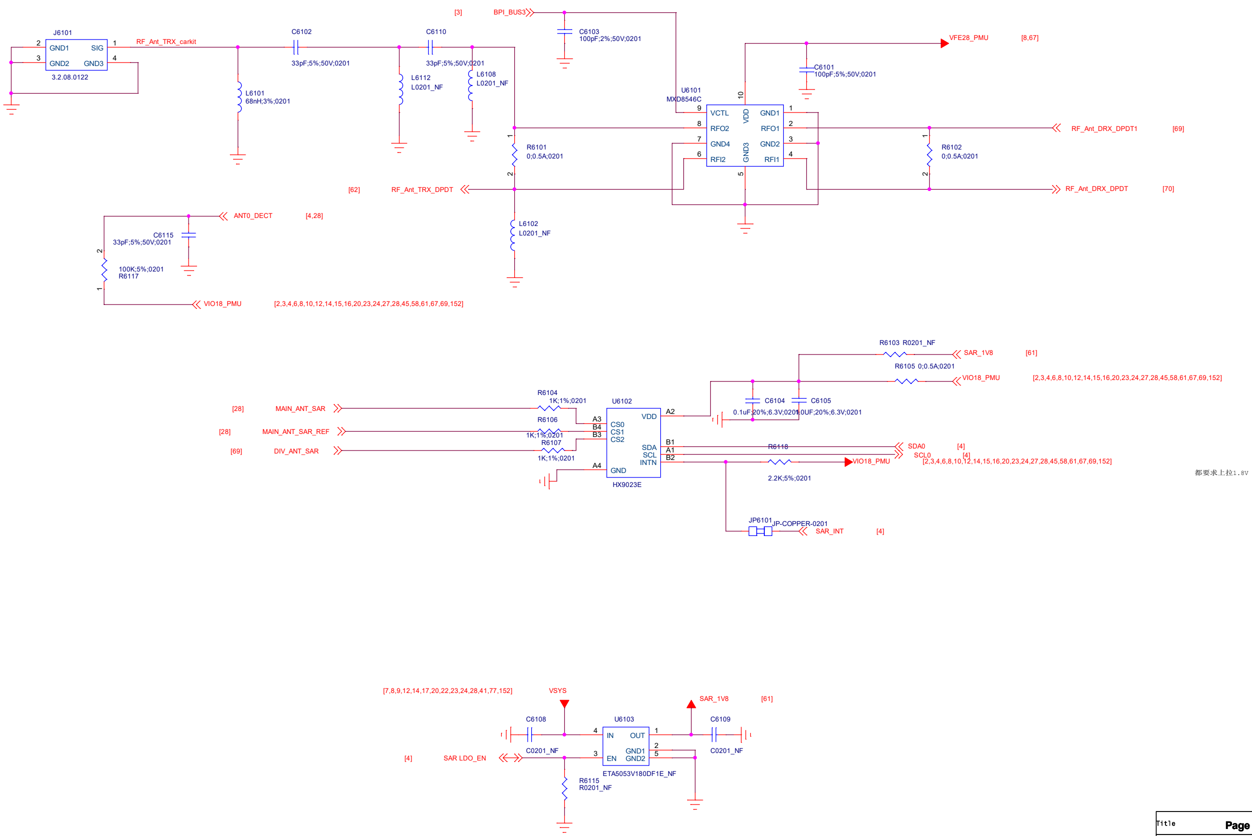


Title	45_SIM/TF IF	REV: V10	
DOCUMENT NO.:	Design Name	Size D	
DEPARTMENT:	WINGTECH-SH	DESIGNER:	liufenglei
Date:	Saturday, May 21, 2022	Sheet 45 of 84	



Title	Page Name = 58_RF_MT6177M
DOCUMENT NO.:	Design Name = N26_1_11M15_20220523
DEPARTMENT:	DEPARTMENT = WINGTECH
DESIGNER:	DESIGNER = LIUFENGLEI
Date:	Page Modify Date = Saturday, May 21, 2022
Sheet	58 of 81

WINGTECH



Title	Page Name = 61_Primary_ANT
REV:	V10
DOCUMENT NO.:	Design Name = N26_1_11M15_20220523_1136
DEPARTMENT:	DEPARTMENT = WINGTECH-SZ
DESIGNER:	DESIGNER = LIUFENGLE
Date:	Page Modify Date = Saturday, May 21, 2022
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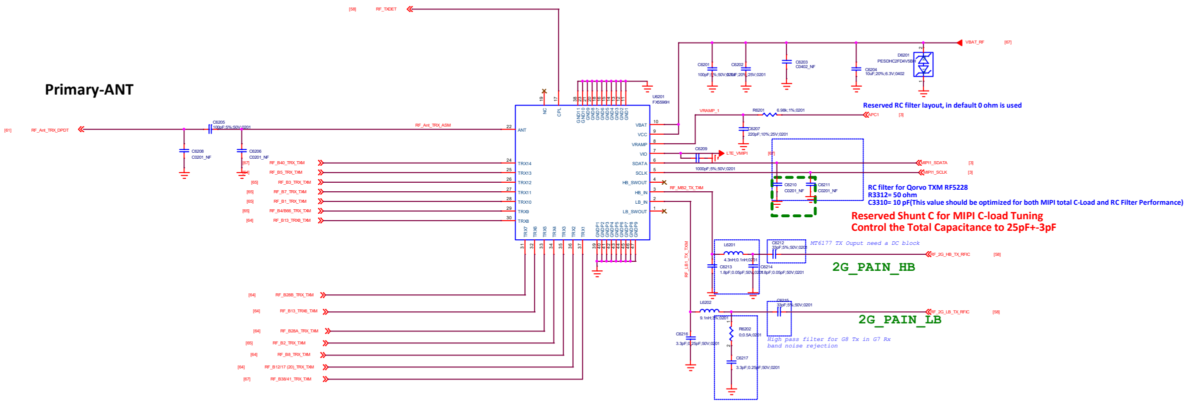
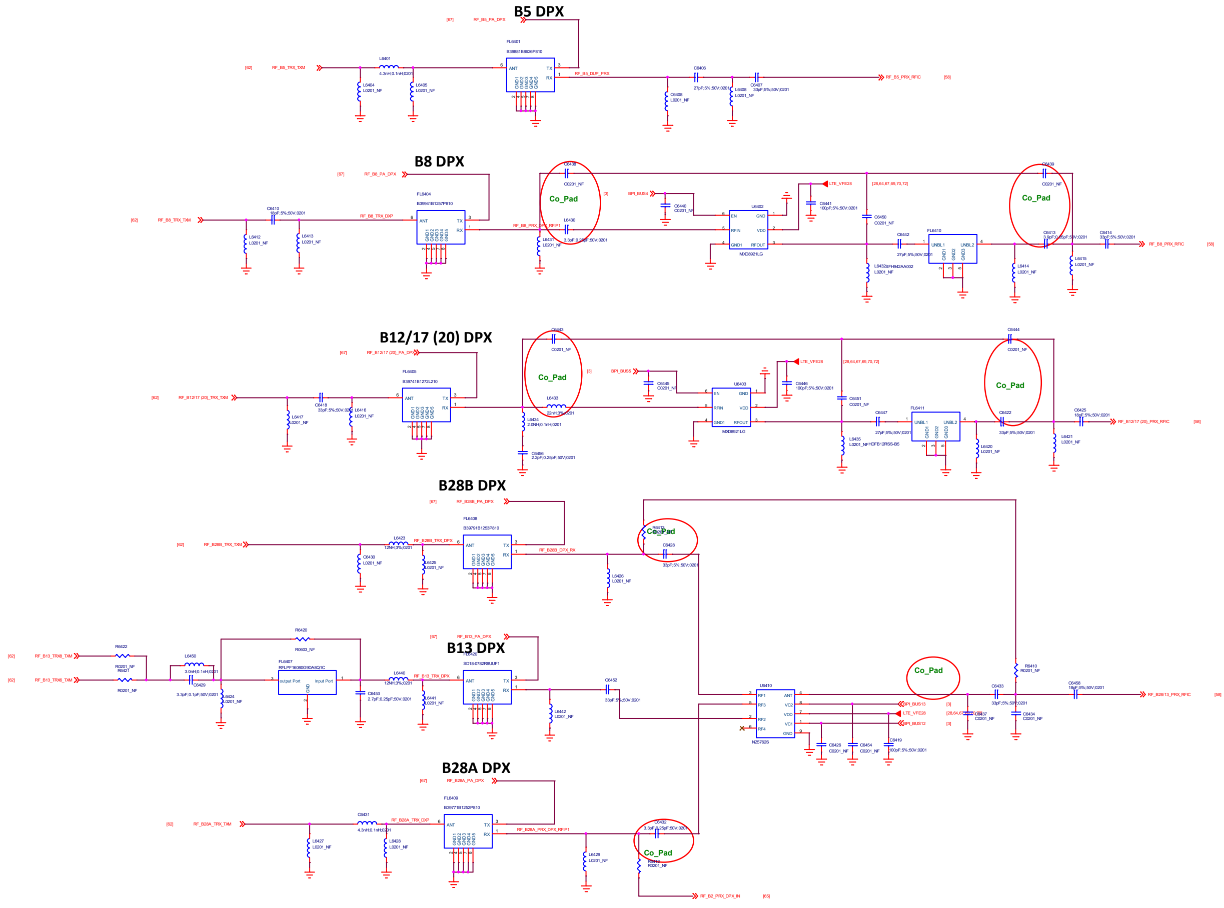
WINGTECH

Primary-ANT



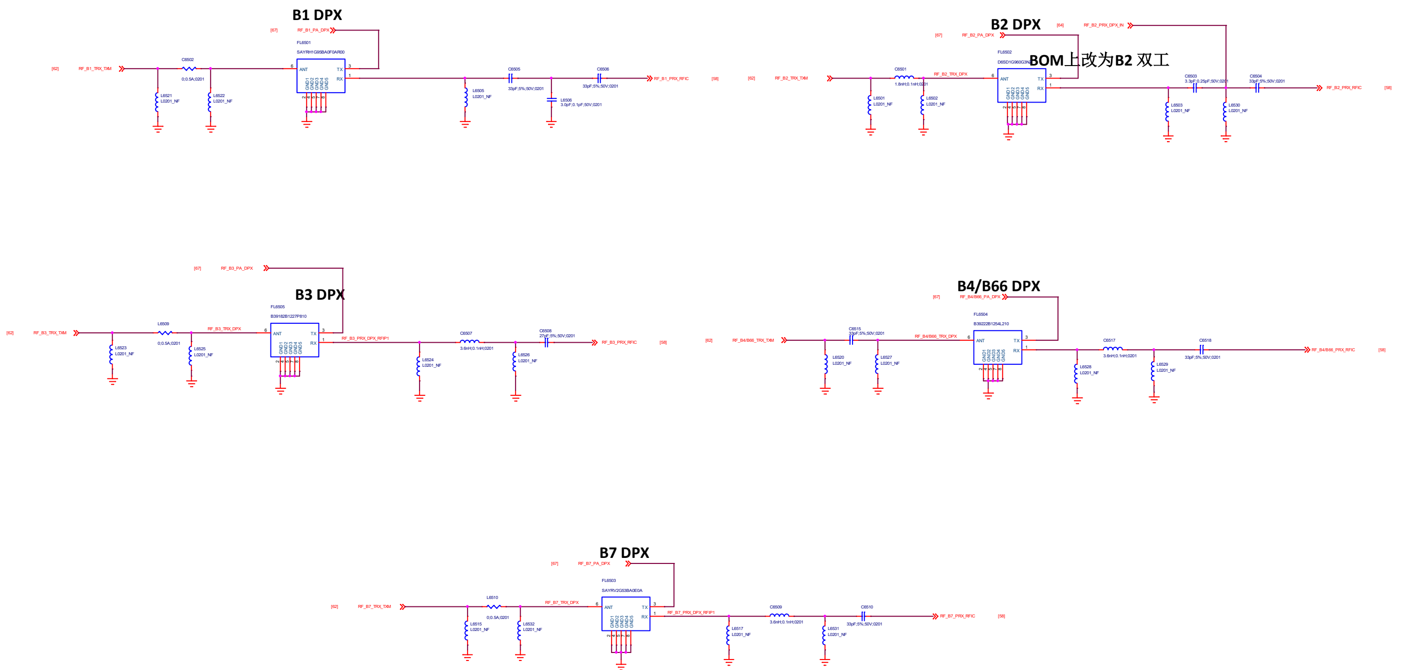
Title	Page Name = 62_Primary_ASM
DOCUMENT NO.:	Design Name = N26_1_11M15_20220523
DEPARTMENT:	DEPARTMENT = WINGTECH
DESIGNER	DESIGNER = LIUFENGLEI
Rev:	V10
Date:	Page Modify Date = Saturday, May 21, 2022
Sheet	62 of 81

WINGTECH

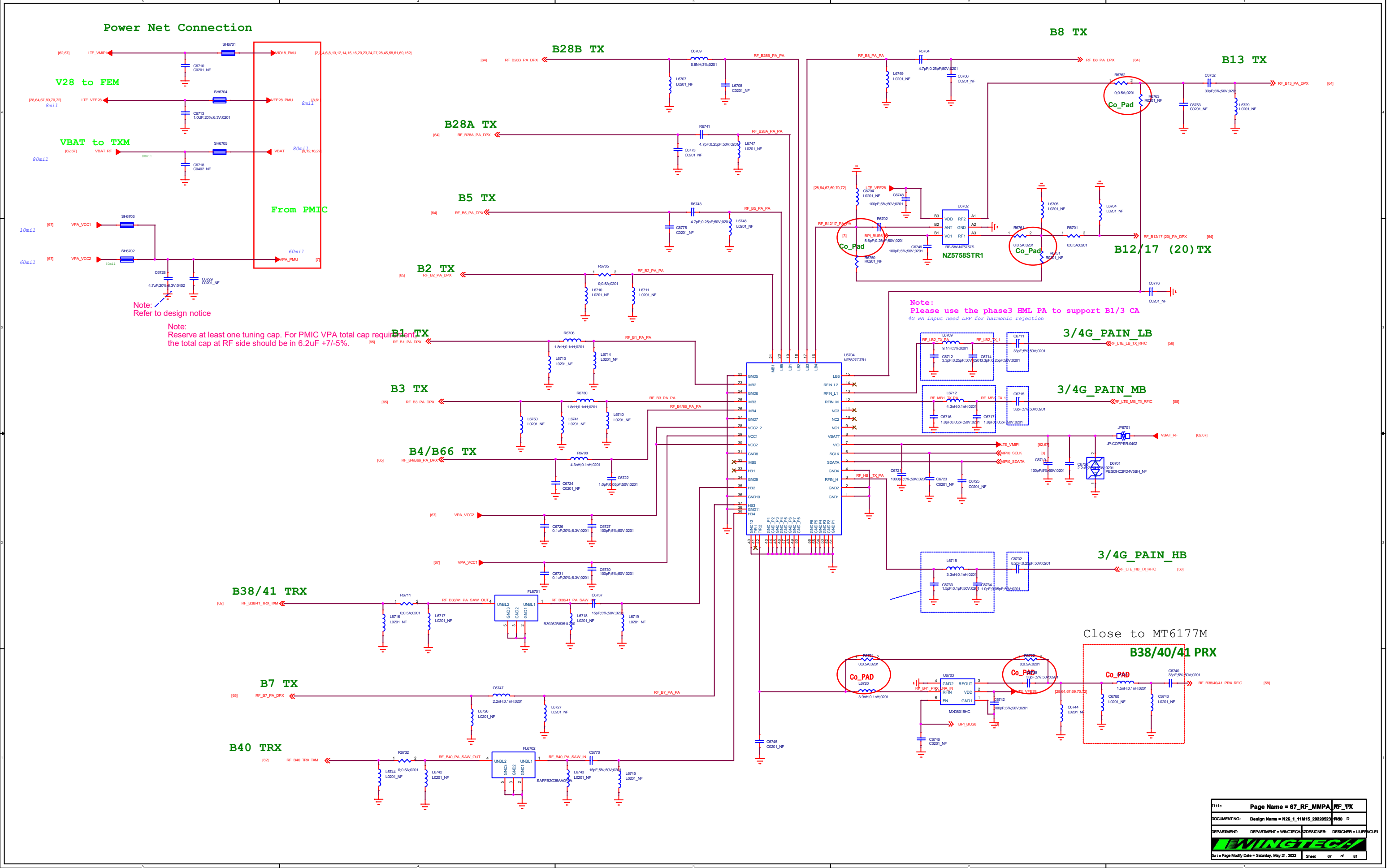


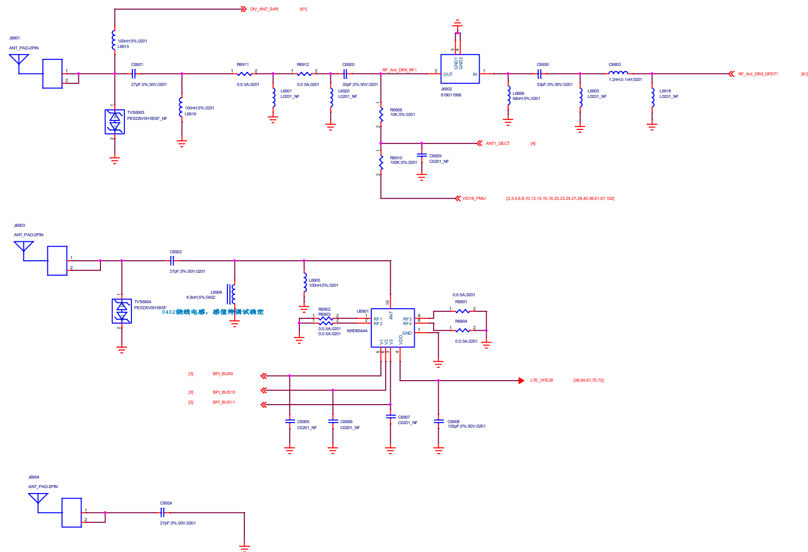
Title	Page Name = 64_TRX_LB	REV: V10
DOCUMENT NO.:	Design Name = N25_1_11M15_20220523	D
DEPARTMENT:	DEPARTMENT = WINGTECH	DESIGNER: DESIGNER = LIUFENGLEI
Date:	Page Modify Date = Saturday, May 21, 2022	Sheet 64 of 81

WINGTECH



Title	Page Name = 65_TRX_MHB	REV: V10	
DOCUMENT NO.:	Design Name = N26_1_11M15_20220523_1180	D	
DEPARTMENT:	DEPARTMENT = WINGTECH-SZ	DESIGNER:	DESIGNER = LIUFENGLEI
Date:	Page Modify Date = Saturday, May 21, 2022	Sheet 65	of 81


Power Net Connection
V28 to FEM
VBAT to TXM
From PMIC
Note:
Refer to design notice
Note:
Reserve at least one tuning cap. For PMIC VPA total cap requirement,
the total cap at RF side should be in 6.2uF +7/-5%.
B28B TX
B28A TX
B5 TX
B2 TX
B1 TX
B3 TX
B4/B66 TX
B38/41 TRX
B7 TX
B40 TRX
B8 TX
B13 TX
B12/17 (20) TX
Co_Pad
NZ5758STR1
Note:
Please use the phase3 HML PA to support B1/3 CA
4G PA input need LPF for harmonic rejection
3/4G_PAIN_LB
3/4G_PAIN_MB
3/4G_PAIN_HB
Close to MT6177M
B38/40/41 PRX
Co_PAD
Page Name = 67_RF_MMPA_RF_TX
DEPARTMENT = WINGTECH-SZ
DESIGNER = LIUFENGLEI
Date Page Modify Date = Saturday, May 21, 2022
Sheet 67 of 81



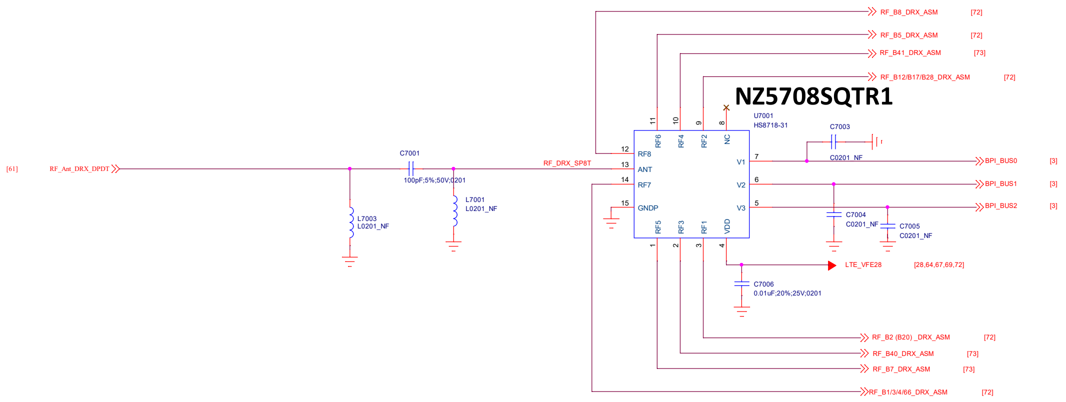
Title Page Name = 69_Diversity_ANT V10

DOCUMENT NO.: Design Name = N26_1_11M15_20220523_1930 D

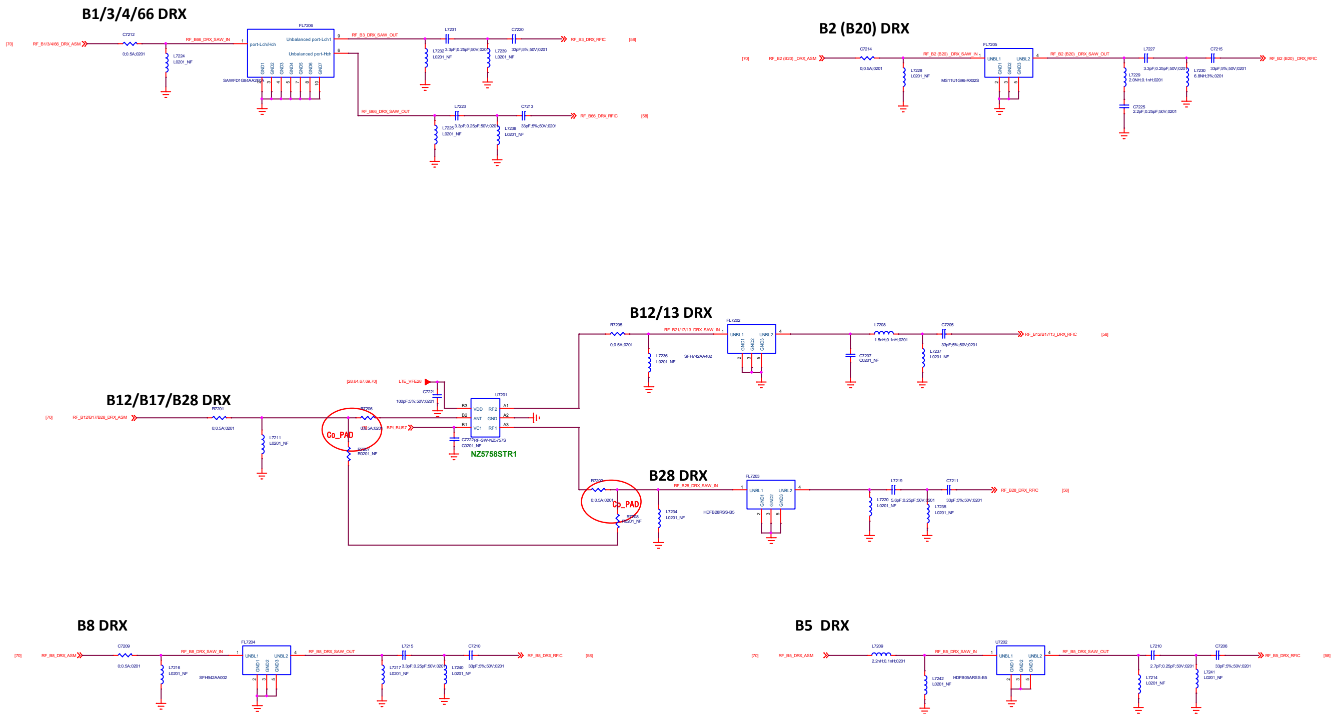
DEPARTMENT: DEPARTMENT = WINGTECH-SZ DESIGNER: DESIGNER = LIUFENGLEI

WINGTECH

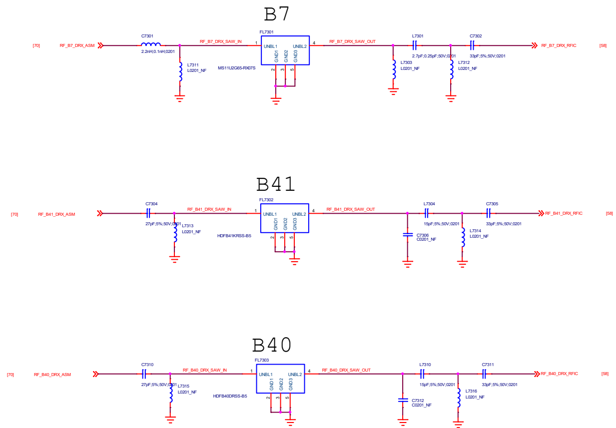
Date: Page Modify Date = Saturday, May 21, 2022 Sheet 69 of 81



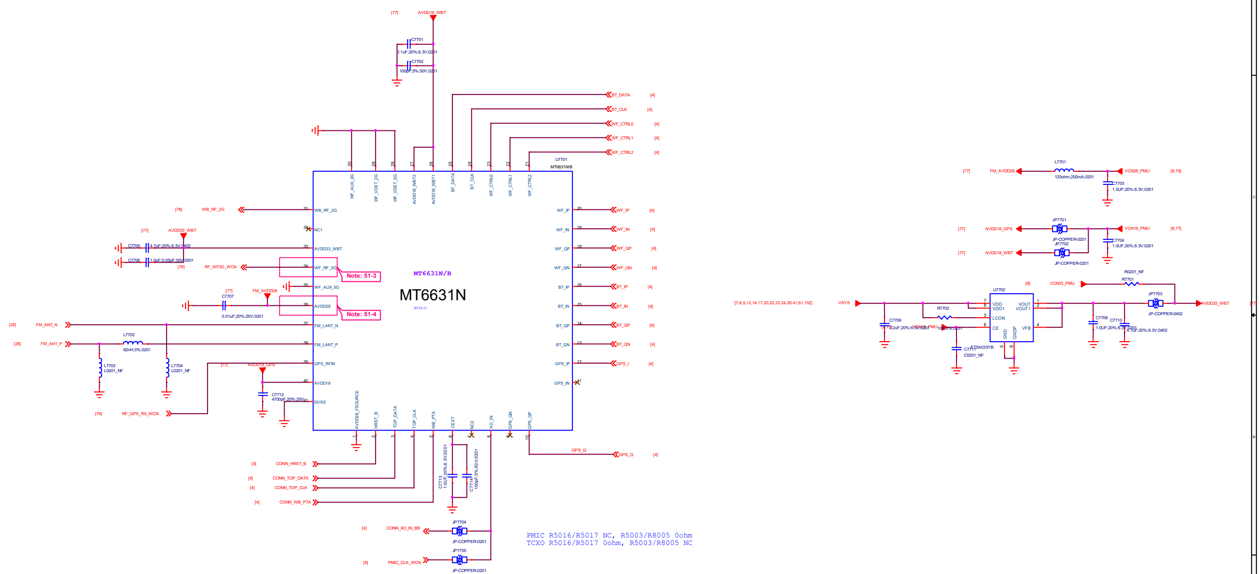
Title	Page Name = 70_Diversity ASM	REV: V10	
DOCUMENT NO.:	Design Name = N26_1_11M15_20220523_1130	Size C	
DEPARTMENT:	DEPARTMENT = WINGTECH-SZ	DESIGNER:	DESIGNER = LIUFENGLEI
Date:	Page Modify Date = Saturday, May 21, 2022	Sheet 70	of 81



Title | Page Name = 72_DRX_LMB | REV: V10
DOCUMENT NO.: | Design Name = N26_1_11M15_20220523_1930 | D
DEPARTMENT: | DEPARTMENT = WINGTECH | DESIGNER: DESIGNER = LIUFENGLEI
WINGTECH
Date: Page Modify Date = Saturday, May 21, 2022 | Sheet 72 of 81



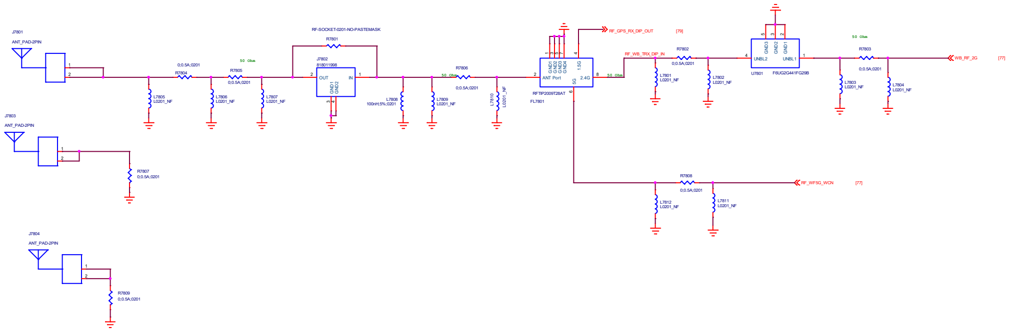
Title	Page Name = 73_DRX_HB	REV: V10
DOCUMENT NO.:	Design Name = N26_1_11M15_20220523_1930	Size D
DEPARTMENT: DEPARTMENT = WINGTECH-SZ	DESIGNER: DESIGNER = LIUFENGLEI	
WINGTECH		
Date: Page Modify Date = Saturday, May 21, 2022	Sheet 73 of 81	



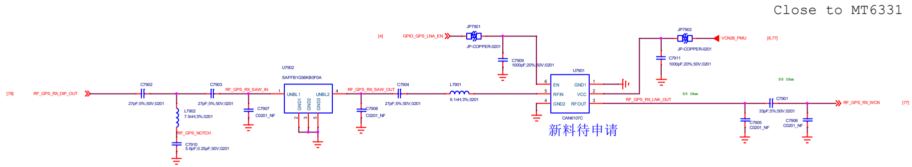
Schematic design notice of "51_CONNECTIVITY_CONSYS_MT6631"	
Note 51-1:	For R5015 size, please select 0402 size or larger one
Note 51-2:	Please refer to MT6762 Baseband design notice for VCN33 LDO selection guide
Note 51-3:	If WiFi 5G not support, connect pin 34(WF_RF_5G) to GND
Note 51-4:	Pin 36 (AVDD28_FM) must be connected to VCN28 even if FM not support
Note 51-5:	If WiFi 5G were no need, VCN33 could be chosen from PMIC output (VCN33_PMU)

Title	Page Name = 77_WCN_MT6631	REV:	V10
DOCUMENT NO.:	Design Name = N26_1_11M15_20220523_1930		D
DEPARTMENT:	DEPARTMENT = WINGTECH-SZ	DESIGNER:	DESIGNER = PENGLIZHONG
Date:	Page Modify Date = Saturday, May 21, 2022	Sheet 77	of 83

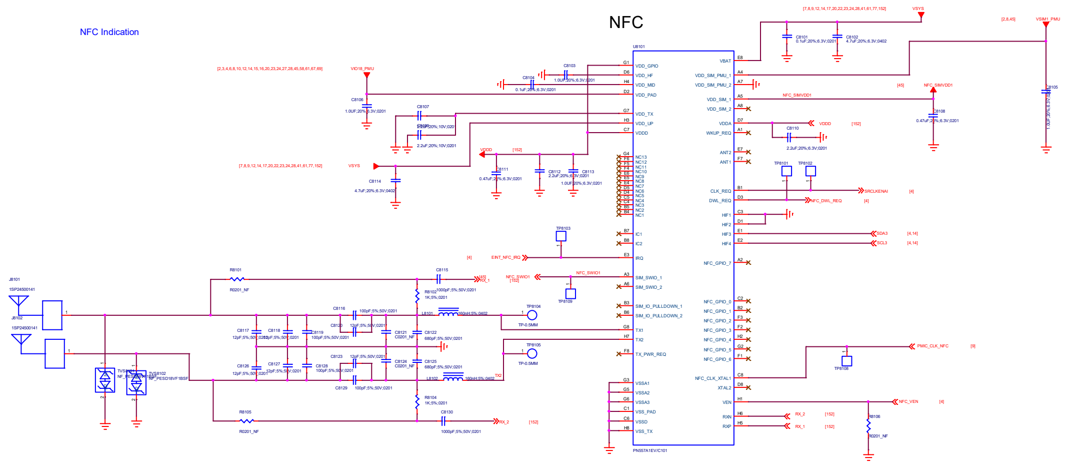
2.4G Wi-Fi BT GPS



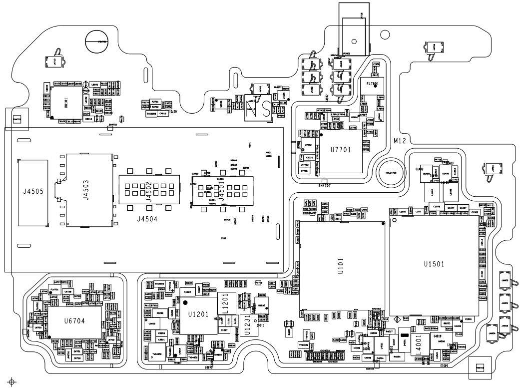
Title	Reserved	REV: V10
DOCUMENT NO.:	Design Name	Size D
DEPARTMENT: WINGTECH-SH		DESIGNER: PENGLIZHONG
Date: Saturday, May 21, 2022		Sheet 78 of 83



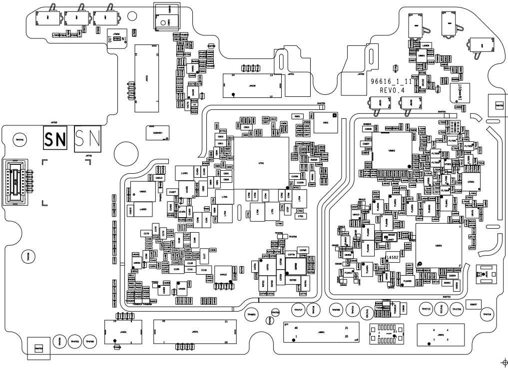
Title	Reserved	REV: V10	
DOCUMENT NO.:	Design Name	Size D	
DEPARTMENT:	WINGTECH-SH	DESIGNER:	PENGLIZHONG
Date:	Saturday, May 21, 2022	Sheet 79 of 83	

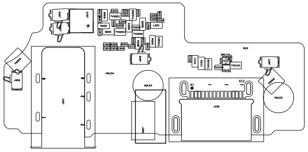


Title	81_NFC_PN557	REV:	V10
DOCUMENT NO.:	[illegible]	Size	D
DEPARTMENT:	WINGTECH-SH	DESIGNER:	liufenglei
Date:	Saturday, May 21, 2022	Sheet	152 of 152

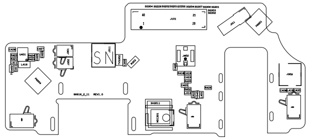


J4505
J4503
J4502
J4504
J4501
U6101
FL7601
U7701
M12
U101
U1501
U1201
L1201
U1231
U6704
L4001


96616_1_11
REV0.4
SN
SN



J304
J310
SN
J203
C201
TVS201
R305
B203
TVS203
L201
TVS204
B204
J405
TVS301
R304
R303
C302
C303
J403
J406
J201
HOLE4
HOLE3
J407
M12
EMI101
TVS104
TVS102
TVS101
TVS103
A1
A12
J102
HOLE2
J308



SG304 SG226 SG202 SG201 SG220 SG204 SG307 SG306 SG301
SG303
SG302
J101
40
21
1
20
J411
C428
U401
R421
R424
R423
L416
C427
J207
J211
J202
R203
J402
R418
C419
C423
R420
C425
R422
L414
R426
J309
C409
L418
C430
96616_2_11 REV1.0
SH3011
MIC301